

Not All Bad Demonstrations Are Equally Bad: Quantifying How Demonstration Failure Modes Degrade Closed-Loop Policy Performance

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<https://github.com/imjbassi/demo-quality-robustness>

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Abstract

We isolate five demonstration defects in a controlled 2D block-pushing task while holding task, dataset size, and training protocol fixed. Across 640 policy fits, ten paired seeds, two policy families, and a transition-matched control, defect cost is not determined by retained primary-expert action information. At full contamination, accidental-success and occlusion data have nearly equal label-fidelity MSE (0.207 and 0.212), yet MLP rollout success is 0.009 and 0.866; inconsistent-strategy data has worse label fidelity (0.475) but higher success (0.229). Thus the robust result is not merely that low-fidelity labels are harmful: accidental success exhibits a late cliff, while the middle ranking changes with policy class. Offline evaluation is similarly non-identifying: 49 policies with open-loop MSE in $[0.20, 0.30]$ span success from 0.130 to 0.995. The alternate controller is exactly Markovian in the recorded 10-D state and its state-only oracle succeeds on every rollout; nevertheless, its MLP clone attains low held-out one-step error (0.0568) but only 0.229 success. This separates observation adequacy from closed-loop clonability and supports defect- and learner-specific quality control rather than a universal data-quality score.

1 Introduction

Behaviour cloning is a standard baseline for robot manipulation [1], and its central known weakness is compounding error: small action errors move the policy off the demonstration distribution, where its predictions degrade further [2]. The standard responses to this are architectural or algorithmic: better policy classes [10], or interactive data collection.

Demonstrations fail in structured ways: a camera is occluded, an operator fumbles into success, strategies disagree, or an episode ends early. These defects differ in whether states, actions, temporal coverage, or strategy are altered. A finite quality-control budget therefore raises a narrower question than whether “better data helps”: after accounting for retained expert action information, do defect type and learner still change the outcome?

1.1 Related work and scope

RoboMimic established that offline human-data quality and the mismatch between training and rollout objectives materially affect robot-learning conclusions [3]. Belkale et al. separately study data quality in imitation learning [9]. Methods for imperfect or suboptimal demonstrations commonly estimate confidence or importance weights [4, 5], or learn discriminator-derived BC weights [6]. Other work critiques and filters heterogeneous demonstrations [7] or retrieves relevant behaviours from larger datasets [8]. These methods ask how to learn despite mixed-quality data. Our smaller contribution is diagnostic: compare named pipeline defects at controlled prevalence, measure their primary-expert label fidelity directly, and test whether offline action error identifies the resulting closed-loop damage. RoboMimic’s observation that training and rollout objectives can select different checkpoints motivates, rather than conflicts with, this controlled comparison [3].

This paper answers both questions in a deliberately small, fully controlled setting. Our contributions are:

1. A controlled design treating demonstration quality as the independent variable across five failure modes, ten paired seeds, and two policy families.
2. A label-fidelity control showing that full-contamination performance is not monotone in disagreement with the primary expert.
3. A similar-offline-score comparison showing that open-loop MSE does not uniquely identify closed-loop success; nested correlations are secondary.
4. Policy-, transition-, and state-only-oracle controls separating defect effects, strategy mixing, observation adequacy, and clonability.

2 Method

2.1 Environment

Push2D is a 2D block-pushing task implemented in vectorised NumPy. A point gripper must move a block into a goal region.

Pushing was chosen over reaching deliberately: it imposes a *staging* requirement, since the gripper must position itself behind the block relative to the goal before pushing is productive. This makes compounding error consequential. A policy that drifts during approach ends up on the wrong side and pushes the block *away* from the goal, which is unrecoverable within the episode budget. This is precisely the class of failure that open-loop metrics are structurally poor at surfacing, since each individual action error is small.

Observations are 10-dimensional (gripper, block and goal positions plus two relative vectors); actions are 2D displacements clipped to $[-1, 1]$. Episodes run for at most 110 steps and terminate on success, defined as the block centre falling within 0.05 of the goal.

2.2 Scripted experts

Using scripted rather than human demonstrations is what makes the experiment controlled: corruption can be applied with exact, programmatic ground truth rather than human judgement about what counts as a bad demo.

The **primary expert** is a hand-coded orbit-then-push controller that circles the block at a safe radius until aligned, then pushes toward the goal. Measured success rate: **100.0%**.

The **alternate expert** always circles in the same rotational direction from a wider staging radius, producing a visibly different route through state space. It was explicitly tuned until it *also* reaches **100.0%**, so that mixing the two tests strategy *inconsistency* rather than strategy *quality*. This tuning turns out to matter substantially (Section 4). Importantly, the controller is a deterministic function of the current 10-D observation; it has no hidden memory or history input.

The **naive controller** charges blindly at the block and never reasons about the goal; the block travels along whatever ray the gripper started on. It succeeds on approximately 6% of episodes, purely by geometric luck.

2.3 Failure modes

Each mode produces demonstrations that a real pipeline could plausibly log:

Occlusion. A contiguous window covering 35% of timesteps has its observations frozen at the last visible frame. Actions remain expert-quality; the demonstrator can see, the camera cannot. The resulting transitions have corrupted inputs and pristine labels.

Accidental success. Rollouts from the naive controller, keeping only those that reached the goal. Correct outcome, no correct behaviour anywhere in the episode.

Corrective flailing. Heavy uniform jitter on the first 12 timesteps, after which the demonstrator settles into correct motion and still succeeds.

Truncation. A clean episode cut at 70% of its length and logged anyway, what a QC filter missing a premature stop looks like.

Inconsistent strategy. Complete episodes from the alternate expert. Every one is a genuine success; the only defect is

disagreement with the primary strategy about what to do in a given state.

2.4 Two design decisions

Action-level corruptions are re-simulated, not pasted.

Overwriting actions in a logged trajectory without re-running the dynamics yields observations that no longer follow from the actions, a physically impossible episode. That measures label noise, not bad demonstrations. Every mode that touches actions replays the episode through the environment.

Corrupted episodes must pass the same success-based QC filter a real pipeline applies. Modes are rejection-sampled to successful episodes only. Truncation is the deliberate exception, since it *is* the filter failing.

2.5 Sweep, policies, evaluation and uncertainty

The sweep axis is **contamination rate** ρ , not per-episode severity. Every dataset contains the same number of episodes; ρ of them are defective and $(1 - \rho)$ are clean, with per-episode severity held fixed. This places all five modes on one comparable axis and answers the question a pipeline actually asks: what fraction of the dataset can be this kind of bad before the policy breaks?

We sweep $\rho \in \{0, 0.25, 0.5, 0.75, 0.9, 1.0\}$ across ten training seeds. The shared clean baseline is fitted once per seed, producing 260 fits per policy family: ten clean fits and $5 \times 5 \times 10$ contaminated fits. The primary policy is a 2×256 ReLU MLP trained by MSE behaviour cloning for 50 epochs with Adam, learning rate 10^{-3} , batch size 256, and no weight decay. A robustness sweep uses an Extra Trees regressor with 40 trees, all input features available at each split, and a minimum leaf size of two [11]. Both train on the same 400-episode datasets. Because episode length varies, these contain 5,957–19,180 transitions across the sweep (clean datasets average 12,207); the transition-matched control below removes this variation.

Because alternate-strategy episodes are longer, we add a control that fixes the dataset at 13,000 transitions and defines ρ as the exact fraction of transitions drawn from that strategy. Repeating all six mixture levels for both policy families and ten seeds adds 120 fits, for 640 fits in total.

Two metrics are recorded per policy. **Open-loop MSE** is action-prediction error against 300 held-out *clean* expert episodes. **Closed-loop success rate** runs the policy over 200 episodes. Evaluation seed $12345 + s$ is used for training seed s : evaluation states vary across seeds but are paired across configurations within a seed. We report means and two-sided 95% Student-t intervals across the ten paired seeds; full-contamination changes are paired against the corresponding clean fit.

Two diagnostics address alternative explanations. **Label fidelity** is the MSE between each logged demonstration action and the primary expert’s action evaluated on that same logged observation, summarized over ten independently generated full-contamination datasets per mode. This quantity measures retained primary-expert action information, not tra-

jectory success. For the alternate strategy, we also evaluate cloned-policy MSE on 300 held-out alternate-expert episodes and compare its rollout success with the original state-only alternate controller on the same evaluation seeds.

3 Results

The MLP clean baseline reaches 0.949 ± 0.015 closed-loop success (mean and 95% t-interval half-width); the Extra Trees baseline reaches 0.996 ± 0.003 .

3.1 The ranking is not explained by label fidelity

Table 1 gives the headline comparison. At full contamination, corrective flailing changes MLP mean success by $+0.016 \pm 0.022$ relative to the paired clean baseline, whereas accidental success changes it by -0.940 ± 0.020 . Both are “bad demonstrations” under a practical taxonomy, yet their observed costs differ sharply.

That contrast alone could be tautological because flailing retains mostly expert actions whereas accidental success does not. The explicit fidelity control rejects a one-dimensional explanation. Accidental-success and occlusion labels have nearly identical disagreement with the primary expert (0.207 versus 0.212 MSE), but their full-contamination success differs by 0.857. Inconsistent-strategy labels disagree most (0.475) yet clone far better than accidental-success data. Truncation has essentially perfect label fidelity but is benign for the MLP and damaging for Extra Trees (Section 4). Performance is therefore not monotone in retained primary-expert action information. The narrower robust claim is the accidental success cliff and a policy-dependent middle ranking.

Corrective flailing causes little degradation in this setting. Jitter-then-recover trajectories may provide DAGger-like coverage of off-distribution states [2]; at $\rho = 1.0$, MLP mean success is 0.965 ± 0.015 . The paired interval includes zero change, so these data do not establish a benefit, but they make flailing a lower QC priority here.

Truncation does not reduce mean success in the observed setting: MLP mean success is 0.962 ± 0.017 at $\rho = 1.0$. A plausible explanation is that the push action can be extrapolated from the retained approach phase. This interpretation should not be transferred to tasks with a distinct terminal phase, such as grasping, without further experiments.

3.2 Accidental success is a cliff, not a slope

The most operationally significant observed result is the shape of the accidental success curve: 0.939 at $\rho = 0.5$, 0.917 at $\rho = 0.75$, 0.666 at $\rho = 0.9$, and **0.009** at $\rho = 1.0$.

In this experiment, a minority of clean demonstrations masks much of the problem. A pipeline monitoring only success-labelled demonstration throughput would not reveal the approaching policy failure. The diagnostic wrong-side push rate tells the same story, rising from 0.011 at $\rho = 0.5$ to 0.415 at

Failure mode	Label MSE	$\rho=0.5$	$\rho=1.0$
Corrective flailing	0.066	0.963	0.965
Truncated episodes	0.000	0.940	0.962
Occlusion (stale obs)	0.212	0.905	0.866
Inconsistent strategy	0.475	0.934	0.229
Accidental success	0.207	0.939	0.009
<i>Clean baseline</i>	0.000	<i>0.949</i>	

Table 1: Primary-expert label-fidelity MSE at full contamination and MLP closed-loop success by contamination rate, means over ten seeds. Label MSE compares logged actions with primary-expert actions on the same logged states.

Failure mode	MSE SD	Success SD	Pearson r
Corrective flailing	0.007	0.030	-0.26
Truncated episodes	0.010	0.029	-0.13
Occlusion	0.026	0.046	-0.64
Inconsistent strategy	0.070	0.312	-0.89
Accidental success	0.117	0.347	-0.95

Table 2: Within-mode sample SDs and descriptive correlations over 60 nested seed-by-contamination fits per mode.

$\rho = 1.0$: the policy has learned to shove the block in essentially arbitrary directions.

3.3 Open-loop evaluation has failure-mode-dependent value

The primary evidence is a similar-score comparison: among the 49 runs whose open-loop MSE lies in the descriptive band $[0.20, 0.30]$, closed-loop success ranges from **0.130** to **0.995**. The band was selected after inspection and is not a statistical equivalence test; it demonstrates non-identification—numerically similar offline scores can accompany nearly the full range of task outcomes. Figure 2 shows the complete distribution.

Correlations provide secondary context. Pooled across all 260 distinct MLP fits, open-loop MSE correlates with success at $r = -0.92$, but this aggregate is driven partly by between-mode separation and high-MSE collapse points. Within-mode coefficients range from -0.13 to -0.95 . Table 2 makes the associated range restriction explicit: truncation and flailing have success SDs near 0.03, so their weak coefficients are not evidence that MSE would fail over a wider outcome range. Each coefficient is descriptive over 60 observations nested within ten seeds and six contamination levels; the points are not independent draws and no correlation significance test is claimed.

4 Robustness controls

4.1 Policy family changes the ranking

Table 3 repeats the full-contamination comparison with Extra Trees. Accidental success remains catastrophic and corrective flailing remains benign. The middle of the ranking changes

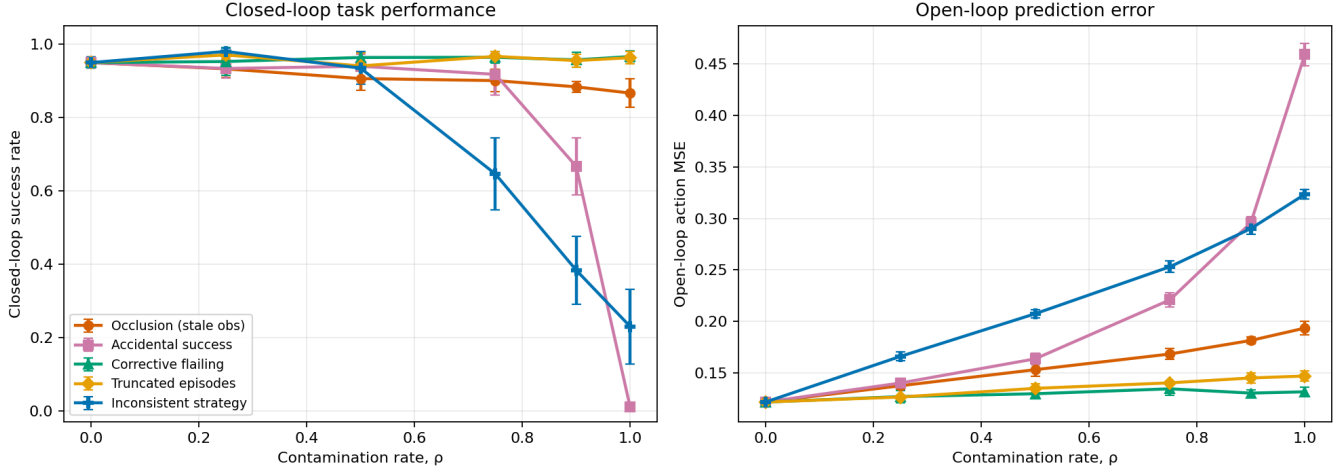


Figure 1: Dose-response curves across contamination rate. **Left:** closed-loop success rate, the quantity of interest. **Right:** open-loop MSE, the offline proxy. Accidental-success policies retain high mean success at $\rho = 0.75$ before collapsing at $\rho = 1.0$, whereas corrective flailing and truncation remain comparatively stable. Points are MLP means and error bars are 95% t intervals over ten paired seeds.

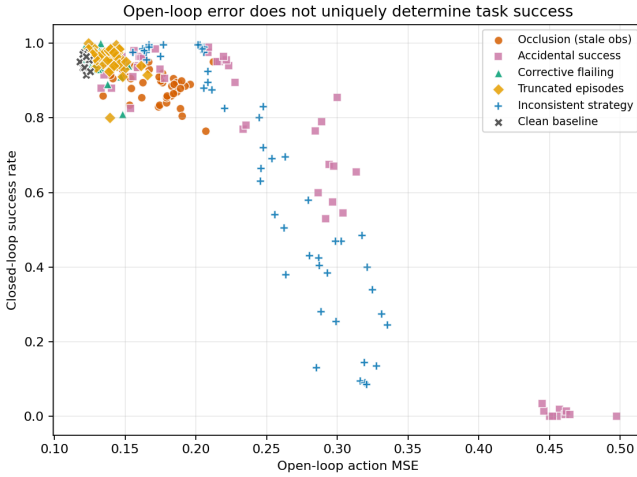


Figure 2: Open-loop MSE against closed-loop success for all 260 distinct MLP fits. Within the descriptive MSE band $[0.20, 0.30]$, closed-loop success spans the range 0.130 to 0.995.

substantially: trees are robust to stale observations but sensitive to truncation, whereas the MLP shows the opposite pattern. Demonstration defects therefore interact with policy inductive bias; only the extreme cases are stable enough here to support a cross-policy priority.

4.2 Strategy mixing and observation adequacy

We initially predicted that mixing two valid strategies would cause multimodal-averaging collapse. The endpoint control contradicts that reading: at $\rho = 1.0$ the dataset is fully self-consistent, so its low success is a *clonability floor*, not evidence of mixing harm. Under episode matching, the MLP’s observed success exceeds the line between its $\rho = 0$ and $\rho = 1$ endpoints at every intermediate mixture.

The stricter transition-matched control fixes both total tran-

Failure mode	MLP	Extra Trees
Corrective flailing	0.965 ± 0.015	0.993 ± 0.007
Truncated episodes	0.962 ± 0.017	0.614 ± 0.033
Occlusion (stale obs)	0.866 ± 0.038	0.990 ± 0.006
Inconsistent strategy	0.229 ± 0.102	0.142 ± 0.050
Accidental success	0.009 ± 0.008	0.002 ± 0.003

Table 3: Closed-loop success at $\rho = 1.0$ (mean $\pm$ 95% t-interval half-width over ten paired seeds).

sitions and the alternate-strategy transition fraction. Its intermediate MLP means are 0.962, 0.966, 0.741, and 0.386; Extra Trees means are 0.984, 0.888, 0.797, and 0.563. Every point remains above the corresponding endpoint interpolation. Across both matching schemes, these curves support unequal clonability, rather than strategy conflict, as the dominant explanation (Figure 3).

The proposed hidden-history explanation for that floor does not hold here: the alternate controller is implemented as a deterministic function of the current 10-D observation. The state-only oracle succeeds on all 2,000 evaluation rollouts across the ten seeds. An MLP trained only on alternate demonstrations also achieves held-out alternate-trajectory MSE 0.0568 ± 0.0041 , yet closed-loop success remains 0.229 ± 0.102 (95% t-interval half-widths). Observation is therefore sufficient for the scripted expert, while one-step regression is not sufficient for a robust clone. This pattern is consistent with compounding error or sensitivity near the controller’s switching geometry, but the present diagnostic does not identify which mechanism dominates.

5 Limitations

Controlled simulation only. Push2D omits 3D geometry, contact uncertainty, visual perception, and hardware latency. The

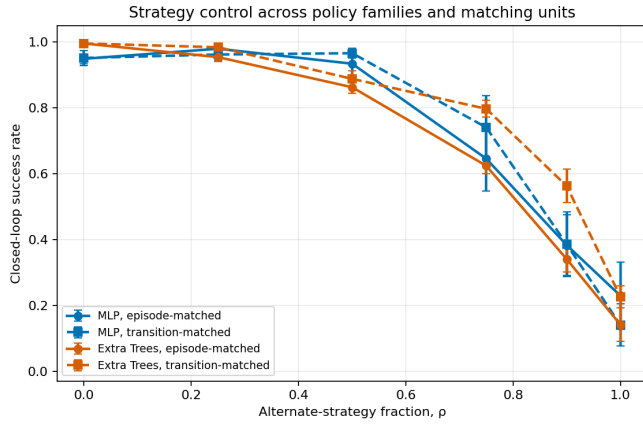


Figure 3: Episode-matched and transition-matched strategy controls. Points are means and error bars are 95% t intervals over ten paired seeds. Every intermediate mixture is above the linearly interpolated endpoint expectation; the low alternate-only endpoint is therefore a clonability floor, not a mixing penalty.

study tests a method for isolating defect effects, not transferable numerical thresholds.

Two deterministic regressors. The MLP and Extra Trees differ in inductive bias, but both return point actions. Multimodal generative policies, including diffusion policies [10], may respond differently to inconsistent strategies.

Finite seed and evaluation coverage. Ten paired seeds materially improve precision, and evaluation states vary by seed, but each fit is still tested on only 200 initial conditions from the same simulator distribution.

Exploratory analysis. The failure modes and MSE band were not preregistered, and intervals are not adjusted for multiple comparisons. They quantify run-to-run uncertainty rather than establish universal effects.

Scripted demonstrations. Programmatic defects provide exact control but do not capture the full diversity or temporal dependence of human data-collection errors.

6 Conclusion

Treating demonstration quality as the independent variable reveals both stable and policy-dependent effects that cannot be reduced to action-label fidelity. Accidental-success and occlusion demonstrations disagree similarly with the primary expert but produce radically different MLP policies; occlusion and truncation also exchange rank across the MLP and Extra Trees. The defensible result is therefore narrower than a universal hierarchy: accidental success has a late cliff, while QC priority in the middle depends on defect structure and the intended learner.

The similar-MSE band is the main evaluation result: nearly equal offline scores coexist with success from 0.130 to 0.995. Correlations are only descriptive because seed and contamination observations are nested and some modes have restricted outcome ranges. Finally, transition matching and a state-only oracle turn an apparent strategy-conflict or observation-aliasing effect into a clonability result. Offline screening remains use-

ful, but it does not substitute for closed-loop validation.

Reproducibility statement

Source code, exact training and evaluation seeds, raw per-run results, and the scripts used to generate all three figures are available at <https://github.com/imjbassi/demo-quality-robustness>. The repository also records the software requirements and commands needed to reproduce all 640 policy fits and rebuild this manuscript.

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